Temporally-Aware Citation Link Prediction in Library and Information Science

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Abstract

The prediction of citation links between scientific papers is a fundamental problem in computational bibliometrics and the sociology of science. While preferential attachment and cumulative advantage explain the macroscopic topology of citation networks, predicting microscopic, pair-level citations requires integrating structural, temporal, semantic, and bibliometric signals in a causally rigorous framework. In this study, we formulate citation link prediction as a binary classification task on two large-scale, independent datasets in Library and Information Science (LIS): 259,220 articles from Dimensions.ai and 168,901 articles from OpenAlex, both spanning 1975–2024.

To prevent temporal data leakage—a pervasive flaw in existing citation prediction literature—we enforce strict temporal causality throughout: all features for a potential citation from paper A (published in year t_A) to paper B are computed exclusively from data available prior to year t_A. We employ a rigorous hard-negative sampling strategy, pairing each positive citation with a non-citation drawn from the same temporal window (± 3 years) and fine-grained disciplinary category, forcing models to learn genuine citation dynamics rather than trivial temporal or topical differences.

We engineer a rich feature set spanning five categories: prestige (year-capped citation count and PageRank), temporal dynamics (citation age and in-degree velocity), bibliographic coupling (common references, Jaccard similarity, co-citation overlap), semantic proximity (TF-IDF cosine similarity), and bibliometric metadata (journal prestige, author productivity). Using

Gradient Boosting, we achieve ROC-AUC of 0.8949 ± 0.0006 (Dimensions) and 0.9762 ± 0.0004 (OpenAlex) under 5-fold stratified cross-validation, with Brier scores of 0.1346 and 0.0871 respectively, confirming well-calibrated probability estimates. McNemar's test confirms statistical superiority of Gradient Boosting over Random Forest ($p < 10^{-13}$ for both datasets). Rolling temporal evaluation across four independent test windows confirms stable performance over two decades.

A dedicated semantic ablation experiment reveals that semantic proximity is the dominant single-feature predictor in OpenAlex (AUC drop of 0.0269 when removed), while temporal gap and co-citation overlap dominate in the purely structural Dimensions setting. We discuss the epistemological implications of these findings, suggesting that the high predictability of citations may reflect the bounded, paradigmatic nature of knowledge diffusion within established disciplinary communities, offering a plausible explanatory reading of the empirical results.

Contents

1 Introduction

The act of citation is the fundamental mechanism through which scientific knowledge is validated, contested, and integrated into the scholarly record [23, 26]. Citations form a vast, directed network that encodes both the intellectual lineage of individual papers and the social structure of scientific communities [13]. From a macroscopic perspective, the evolution of citation networks is well-described by generative models such as preferential attachment [2, 10]: highly cited papers accumulate new citations at a rate proportional to their existing prestige, generating the characteristic power-law distribution of citation counts observed across all scientific fields. However, while these models successfully reproduce aggregate network topology, they are insufficient for microscopic prediction: given a newly published paper A and a previously published paper B, what is the probability that A will cite B?

This pair-level prediction task is central to both the sociology of science and the design of modern scientific recommender systems [3, 30]. The challenge lies in the fact that citations are not purely random nor purely prestige-driven; they are deliberate acts of intellectual positioning constrained by temporal relevance, topical proximity, and social structure [6, 17]. A researcher choosing which papers to cite does not merely select the most highly cited works in a field; they select works that are relevant to their specific research question, known to them through their social and institutional networks, and established as credible within their disciplinary community [9].

In recent years, machine learning approaches have increasingly been applied to citation link prediction. However, as noted by several critics, many such models suffer from subtle methodological flaws that artificially inflate performance [32]. The two most pervasive issues are *temporal data leakage*—using future information, such as the total static citation count of paper B, to predict a citation from paper A published years earlier—and *trivial negative sampling*—pairing a recent paper with a random, entirely unrelated paper from decades ago or from a different field, making the classification task artificially easy. These flaws are not merely technical inconveniences; they undermine the validity of any theoretical claims derived from inflated performance metrics.

In this study, we address these methodological challenges directly. We present a strictly temporally-aware machine learning pipeline for predicting citation links within the domain of Library and Information Science (LIS), evaluated on two comprehensive and independent datasets derived from Dimensions.ai and OpenAlex. LIS is an ideal testbed for this investigation: it is a mature, interdisciplinary field that has itself contributed substantially to bibliometric methodology, yet its own citation dynamics have received comparatively less attention than the physical or biomedical sciences. Its breadth—spanning information retrieval, knowledge organization, scholarly communication, and digital libraries—makes it a rich and heterogeneous corpus for studying citation behavior across diverse research traditions.

Our contributions are fourfold. First, we introduce a rigorous, temporally-aware evaluation methodology that substantially mitigates data leakage by capping all network and prestige features strictly prior to the citing paper's publication year. Second, we enrich the standard bibliometric feature set with genuine graph-theoretic features (year-capped PageRank, in-degree velocity) and bibliometric metadata (journal prestige, author productivity), demonstrating their incremental predictive value via leave-one-out ablation. Third, we conduct a dedicated semantic ablation experiment to quantify the independent contribution of semantic proximity and to directly address the concern that high performance in semantically-constrained settings may reflect topical redundancy rather than genuine citation dynamics. Fourth, we discuss the epistemological implications of high citation predictability for the sociology of science, connecting our empirical findings to broader debates about the nature of knowledge diffusion, disciplinary closure, and the structural determination of scientific attention.

The remainder of this paper is organized as follows. Section 2 reviews the relevant literature. Section 3 describes our data sources, feature engineering, and experimental setup. Sections 4 and 5 present results for the Dimensions and OpenAlex datasets. Section 6 provides a comparative analysis. Section 7 discusses theoretical and methodological implications. Section 8 concludes.

2 Literature Review

2.1 Classical Theories of Citation Dynamics

The study of citation patterns is a cornerstone of bibliometrics and the science of science. The earliest and most enduring model of citation dynamics is Merton's concept of the Matthew Effect [23]: the tendency for already recognized scientists and papers to receive disproportionate future recognition, generating a "rich-get-richer" dynamic. De Solla Price formalized this as cumulative advantage in citation networks [11], and Barabási and Albert later generalized it as preferential attachment—a universal mechanism for the emergence of scale-free networks [2]. These models provide a compelling account of the aggregate statistical properties of citation networks but are insufficient for pair-level prediction.

Subsequent research has demonstrated that the strength of preferential attachment varies significantly across disciplines and over time [33]. Bianco and Gabrielli introduced the concept of paper "fitness," suggesting that intrinsic quality or relevance modulates the rich-get-richer effect [4]. Wang, Song, and Barabási demonstrated that the long-term scientific impact of a paper is governed by a combination of initial fitness and cumulative advantage, challenging purely prestige-based models [32]. These theoretical developments motivate our multi-feature approach: citation behavior is not a simple function of prestige but emerges from the complex interaction of prestige, relevance, and social structure.

2.2 Social and Semantic Proximity in Citation

Beyond prestige, a growing body of research has demonstrated that social proximity and semantic proximity are powerful predictors of citation behavior, often rivaling or exceeding the effect of prestige for the majority of papers. Newman's foundational work on co-authorship networks established that scientific collaboration is highly clustered and that co-authorship strongly predicts future collaboration and citation [25]. Kumar extended this analysis to show that citation probability decays rapidly with social distance in co-authorship networks [18]. Martin et al. demonstrated that the social structure of scientific communities—operationalized through co-authorship, institutional affiliation, and conference participation—is a major determinant of citation patterns [22].

The landmark "Citation proximus" study by Kozlowski et al. [16] provided rigorous statistical evidence for the joint role of social and semantic proximity using Generalized Linear Mixed Models, demonstrating that these factors together explain a substantial fraction of the variance in citation behavior across multiple scientific fields. This study opened the door for more computationally advanced, predictive inquiry: the relationships between network structure, semantic content, and prestige are unlikely to be simple and additive; they involve complex, non-linear interactions that traditional statistical models may fail to capture fully.

Semantic proximity has been operationalized through a range of methods, from keyword overlap and topic models [5] to deep neural embeddings [12, 29]. While contextual embeddings from models such as BERT and SBERT offer greater expressiveness, they introduce a potential leakage concern in citation prediction: pretrained language models may have been exposed to the target corpus during their training phase, potentially inflating similarity scores for papers that appear in their training data. We address this concern by using TF-IDF cosine similarity, computed solely from the corpus at hand, which is free from this concern while remaining a well-validated measure of topical proximity.

2.3 Temporal Dynamics and Citation Obsolescence

A dimension often underappreciated in citation prediction is the temporal decay of citation probability. Price's early work on citation half-lives established that the probability of a paper being cited declines with age, reflecting both the natural obsolescence of older work and the tendency for researchers to prioritize recent literature [11]. Redner's analysis of citation distributions confirmed that temporal decay is highly non-uniform: foundational papers accumulate citations over decades, while applied or methodological papers may be superseded within a few years [28]. The temporal gap between citing and cited paper is therefore a complex predictor that interacts with prestige, field, and the nature of the contribution.

Temporal link prediction—predicting which edges will appear in a network at future time steps—is a well-studied problem in network science [19]. In citation networks, temporal prediction is particularly challenging because the network is acyclic (a paper can only cite papers published before it) and because the candidate space grows quadratically with corpus size. Shibata et al. applied temporal link prediction methods to citation networks, finding that structural features alone can achieve reasonable predictive performance but that temporal constraints are essential for valid evaluation [30]. Our rolling temporal evaluation framework directly addresses this challenge by training on papers published before a cutoff year and testing on papers published in a subsequent window, providing a rigorous assessment of temporal generalizability.

2.4 Bibliographic Coupling and Co-citation Analysis

Beyond prestige and temporal dynamics, the structural overlap between reference lists provides a powerful signal for citation prediction. Bibliographic coupling [15]—the number of references shared by two papers—reflects shared intellectual heritage. Co-citation analysis [31]—measuring how often two papers are cited together by a third paper—captures a complementary dimension of intellectual proximity. Together, these reference overlap features capture the bibliographic embedding of a paper in the existing literature, a dimension of proximity that is distinct from both semantic content and social network position.

The Jaccard coefficient of reference sets provides a normalized measure of bibliographic coupling that is invariant to differences in reference list length. Common citers (the number of papers that cite both the citing and cited paper, excluding the citing paper itself) operationalize co-citation overlap in a temporally causal manner. We note that for recently published papers with few citations, co-citation overlap will be sparse or zero; this is an inherent property of the measure rather than a methodological artifact, and it is captured implicitly by the model through its interaction with prestige and temporal gap features.

2.5 Machine Learning for Citation Prediction

The integration of machine learning into bibliometrics has shifted the focus from explanatory models to predictive frameworks. Early predictive efforts employed standard regression models to predict citation counts [20]. More recent studies have leveraged ensemble methods such as Gradient Boosting [24] and Random Forests [7], which are particularly adept at capturing the non-linear interactions between disparate feature types. Chakraborty et al. proposed a stratified learning approach for citation count prediction, demonstrating that models trained on subgroups outperform models trained on the full dataset [8]. Bhagavatula et al. applied content-based citation recommendation using neural embeddings, demonstrating the value of semantic features for citation prediction [3]. Alohali et al. provided a recent systematic review of machine learning approaches to citation analysis [1].

The interpretability of machine learning models is a growing concern in bibliometrics. SHAP (SHapley Additive exPlanations) [21] provides a theoretically grounded framework for attributing model predictions to individual features, enabling researchers to identify which factors drive citation decisions in a given corpus. We employ SHAP analysis throughout this study to bridge the gap between predictive performance and theoretical interpretation.

2.6 Bibliographic Data Infrastructure

Dimensions.ai [14] is a comprehensive, linked research information system integrating publications, citations, grants, patents, and policy documents. Its coverage of the scientific literature is among the broadest available, and it provides structured access to metadata including author affiliations, reference lists, and Field of Research (FoR) classifications based on the ANZSRC 2020 taxonomy.

OpenAlex [27] is a fully open, freely accessible index of scholarly works developed as a successor to Microsoft Academic Graph. OpenAlex is particularly notable for its open data philosophy and its rich topic taxonomy, which enables fine-grained subject classification. The use of two independent data sources in this study is not merely a robustness check; it is a substantive methodological contribution, as findings consistent across Dimensions and OpenAlex are more likely to reflect genuine properties of LIS citation dynamics rather than artifacts of a specific data infrastructure.

3 Data and Methodology

3.1 Corpus Construction

3.1.1 Dimensions.ai Dataset

The Dimensions corpus was constructed by querying the Dimensions.ai API for all journal articles in Library and Information Science published between 1975 and 2024. The retrieval strategy combined two complementary approaches to maximize coverage while maintaining topical relevance.

The primary query filtered by Field of Research (FoR) classification code `4610` (Library and Information Studies) under the ANZSRC 2020 taxonomy, executed year by year from 1975 to 2024:

```
search publications
where category_for.id = "4610"
and year = {Y}
and type = "article"
return publications[id+year+title+authors+journal+
  abstract+times_cited+reference_ids+category_for+doi]
```

To supplement this with articles that may have been misclassified or assigned to adjacent FoR codes, four additional keyword queries were executed against titles and abstracts for each year:

```
search publications in title_abstract_only
for "{keyword}"
where year = {Y} and type = "article"
```

where `keyword` $\in$ {*knowledge organization*, *digital libraries*, *information literacy*, *academic libraries*}. These keywords were selected to capture the major LIS subdisciplines most likely to appear in adjacent FoR categories (education, computer science). Results were deduplicated by Dimensions article ID. The retrieved fields included: unique identifier, publication year, full title, author list with Dimensions author IDs, journal name, abstract text, citation count, reference ID list, FoR classification codes, and DOI. The final Dimensions corpus comprises **259,220 articles**, of which 96.3% have abstracts available.

3.1.2 OpenAlex Dataset

The OpenAlex corpus was constructed using the OpenAlex REST API, filtering by five specific `primary_topic.id` values corresponding to core LIS and scientometrics research areas. The legacy OpenAlex concept `C136764020` ("Library and Information Science") was deliberately avoided, as it is assigned by an automated ML tagger and returns approximately 4.2 million records—far too broad for a focused citation dynamics study. The five topic IDs used were:

Topic ID	Description
`T10712`	Library Science and Information Literacy
`T14330`	Library Science and Information Systems
`T13166`	Information Science and Libraries
`T13673`	Library Science and Information
`T10102`	Scientometrics and Bibliometrics Research

The filter applied OR logic across all five topic IDs, restricted to article type and publication years 1975–2024. The API was queried using the polite pool with an institutional email address. After deduplication by OpenAlex work ID, the final OpenAlex corpus comprises **168,901 articles**, of which 64.3% have abstracts available.

3.1.3 Corpus Characteristics

Table 1: Corpus Characteristics

Characteristic	Dimensions	OpenAlex
Total articles	259,220	168,901
Year range	1975–2024	1975–2024
Articles with abstracts	249,751 (96.3%)	108,611 (64.3%)
Mean citations per article	20.98	6.26
Median citations per article	4	0
Within-corpus citation pairs	331,047	239,349

3.2 Citation Pair Construction and Hard Negative Sampling

The fundamental task is binary classification: given a pair of papers (A, B) where $t_B \leq t_A$, predict whether A cites B. We extracted all observed within-corpus citations where both the citing paper A and the cited paper B exist in our dataset, yielding 331,047 positive pairs for Dimensions and 239,349 for OpenAlex.

For each positive pair (A, B), we generated a hard negative pair (A, B') by sampling a paper B' that satisfies all of the following constraints:

1. $|t_{B'} - t_B| \leq 3$ (temporal plausibility);
2. B' shares at least one Field of Research code with B (Dimensions) or belongs to the same primary topic cluster (OpenAlex) (topical comparability);

3. A does not actually cite B' (verified against the full reference list);

4. $B' \neq B$.

This hard negative sampling strategy is substantially more demanding than uniform random sampling. By constraining negatives to the same temporal neighborhood and topical domain as the positive cited paper, we ensure that the classification task cannot be solved by trivially exploiting temporal or topical differences. The resulting datasets contain **662,094 balanced pairs** for Dimensions and **478,698 balanced pairs** for OpenAlex.

3.3 Temporally-Aware Feature Engineering

We engineer features in five categories. All features that depend on the state of the literature are computed using a strictly causal rolling window: for any citing paper published in year t, every feature is derived solely from papers published up to year $t-1$.

3.3.1 Prestige Features

Prestige of Cited Paper (P_{cited}): The log-transformed cumulative citation count received by paper B from papers published up to year $t_A - 1$:

$$P_{cited}(B, t_A) = \log\left(1 + \sum_{C\,:\,t_C \leq t_A - 1} \mathbf{1}[C \text{ cites } B]\right) \tag{1}$$

This is the primary operationalization of the Matthew Effect: papers that have already attracted many citations are more likely to attract future ones.

Activity of Citing Paper (A_{citing}): The log-transformed cumulative citation count of paper A up to year $t_A - 1$ (OpenAlex only). This captures the prestige of the citing paper itself.

3.3.2 Graph-Theoretic Network Features

Year-Capped PageRank (PR_{cited}): The PageRank centrality of paper B in the directed citation graph G_{t_A-1}, computed using the power-iteration algorithm with damping factor $\alpha = 0.85$:

$$PR(B) = \frac{1-\alpha}{|V|} + \alpha \sum_{C \to B} \frac{PR(C)}{d^+(C)} \tag{2}$$

where $d^+(C)$ is the out-degree of paper C in G_{t_A-1}. Unlike raw citation count, PageRank accounts for the quality of citing papers.

In-degree Velocity (V_{cited}): The proportion of paper B's total citations (up to $t_A - 1$) received in the three years immediately preceding t_A:

$$V_{cited}(B, t_A) = \frac{\sum_{C:\, t_A-3 \le t_C \le t_A-1} \mathbf{1}[C \text{ cites } B]}{\max(1,\ P_{cited}(B, t_A))} \tag{3}$$

This captures the momentum of a paper's citation trajectory, distinguishing papers experiencing a recent surge from those with steady long-term impact.

3.3.3 Bibliographic Coupling and Co-citation Features

Common References (CR): The number of papers cited by both A and B: $CR(A, B) = |R_A \cap R_B|$.

Jaccard Reference Similarity (J_{refs}): The Jaccard coefficient of the reference sets: $J_{refs}(A, B) = |R_A \cap R_B|/|R_A \cup R_B|$.

Co-citation Overlap (CC): The number of papers (other than A) published up to $t_A - 1$ that cite both A and B:

$$CC(A, B) = |\{C \ne A : C \text{ cites } A \text{ and } C \text{ cites } B,\ t_C \le t_A - 1\}| \tag{4}$$

3.3.4 Temporal Feature

Temporal Gap (Δt): $\Delta t(A, B) = t_A - t_B$. This reflects the natural decay of citation probability with age and the tendency for recent papers to cite other recent papers.

3.3.5 Semantic Proximity

Semantic Similarity (S_{sem}): The TF-IDF cosine similarity between the concatenated title and abstract of papers A and B. We fit a TF-IDF vectorizer with up to 50,000 features and bigrams on the full corpus, then compute pairwise cosine similarities for all citation pairs. We explicitly chose TF-IDF over pre-trained neural embeddings (e.g., SBERT) to guarantee no temporal data leakage: pre-trained language models may have observed the target corpus during their training phase, potentially inflating similarity scores. TF-IDF, computed solely from the corpus at hand, is free from this concern.

3.3.6 Bibliometric Metadata Features (Dimensions Only)

Journal Prestige (JP): The total citation count accumulated by all papers published in the same journal as paper B, up to year $t_A - 1$.

Author Productivity (AP): The total number of papers published by the most prolific author of paper A up to year $t_A - 1$.

3.4 Feature Summary

Table 2: Complete Feature Set

Category	**Feature**	**Dimensions**	**OpenAlex**
Prestige	P_{cited} (log citation count)	✓	✓
Prestige	A_{citing} (log citing activity)	–	✓
Graph	PR_{cited} (year-capped PageRank)	✓	✓
Graph	V_{cited} (in-degree velocity)	✓	✓
Bibliographic	CR (common references)	✓	✓
Bibliographic	J_{refs} (Jaccard reference sim.)	✓	✓
Bibliographic	CC (co-citation overlap)	✓	✓
Temporal	Δt (temporal gap)	✓	✓
Semantic	S_{sem} (TF-IDF cosine sim.)	✓	✓
Metadata	JP (journal prestige)	✓	–
Metadata	AP (author productivity)	✓	–
Total		**9**	**9**

3.5 Machine Learning Models

We evaluate four models spanning the linear-to-non-linear spectrum:

Logistic Regression (LR): A linear baseline that models the log-odds of citation as a linear function of features. It provides a transparent benchmark against which to evaluate the added value of non-linear models.

Linear SVM: A linear support vector machine that finds the maximum-margin hyperplane separating cited from non-cited pairs, optimizing hinge loss.

Random Forest (RF): An ensemble of 200 decision trees trained on bootstrap samples [7]. At each node, a random subset of $\sqrt{d}$ features is considered for splitting.

Gradient Boosting (GB): A sequential ensemble method that builds trees to correct the errors of previous trees [24]. We use 200 trees with learning rate 0.1 and maximum depth 5.

3.6 Evaluation Framework

All models are evaluated using **5-fold stratified cross-validation**, preserving the class distribution in each fold. We report mean and standard deviation across folds for six metrics: ROC-AUC, PR-AUC, F1-score, Accuracy, MCC, and Brier score. Additionally, we conduct:

- **Rolling temporal evaluation** across four independent test windows (train on papers before window start, test on papers within window);
- **Temporal hold-out** (train $\leq$ 2015, test 2016–2020);
- **McNemar's test** comparing RF and GB predictions on the full dataset;
- **Calibration analysis** using Brier score and reliability diagram;
- **SHAP analysis** and leave-one-out ablation for the best model;
- **Semantic ablation experiment** (OpenAlex only): full CV with and without semantic similarity for all four models.

4 Results: Dimensions.ai Analysis

4.1 Corpus Overview

Figure 1 shows the temporal distribution of LIS publications in the Dimensions dataset. The corpus exhibits steady growth from the late 1980s through the 2010s, reflecting both the expansion of LIS as a field and the increasing coverage of Dimensions over time. The distribution of citation counts is highly skewed, with a small number of highly cited papers and a large majority with few citations—a characteristic signature of preferential attachment and the Matthew Effect.

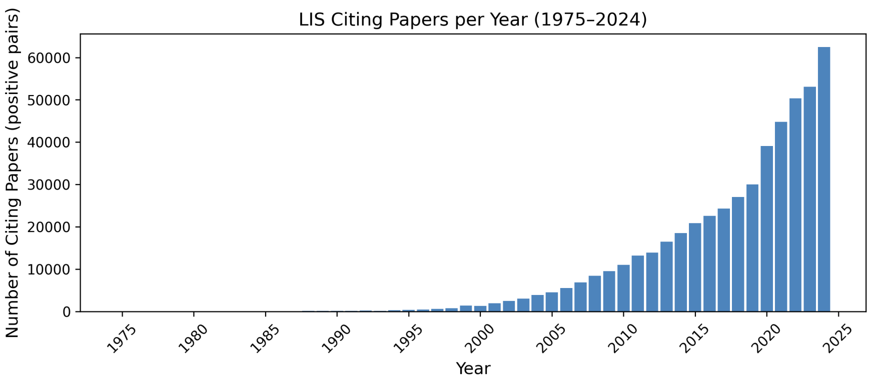

Figure 1: Number of LIS publications per year in the Dimensions dataset (1975–2024).

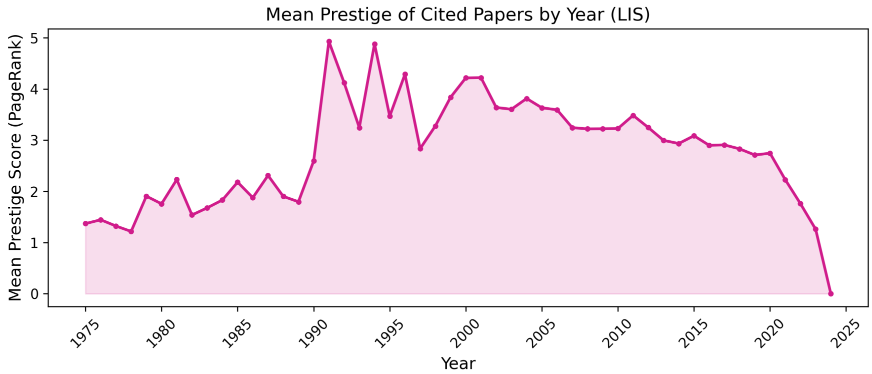

Figure 2: Mean citations per publication year in the Dimensions dataset.

Figure 2 shows the mean citation count per publication year. Papers published in the 1990s and early 2000s have accumulated the highest mean citation counts, reflecting the combined effect of their age (more time to accumulate citations) and their foundational role in establishing the modern LIS research agenda. The decline in mean citations for more recent papers is expected, as these papers have had less time to accumulate citations.

4.2 Model Performance

Table 3 presents the 5-fold stratified cross-validation results for all four models on the Dimensions dataset (662,094 pairs, 9 features). Gradient Boosting achieves the highest performance across all metrics, with ROC-AUC of 0.8949 ± 0.0006, F1 of 0.8013 ± 0.0009, and MCC of 0.6175 ± 0.0016. The Brier score of 0.1346 confirms well-calibrated probability estimates. McNemar's test confirms that Gradient Boosting is statistically superior to Random Forest ($p = 7.20e - 26$).

Table 3: 5-Fold CV Results: Dimensions Dataset (Mean ± Std)

Model	**ROC-AUC**	**PR-AUC**	**F1**	**Accuracy**	**MCC**	**Brier**
Logistic Regression	0.8719 ± 0.0006	0.8893 ± 0.0008	0.7757 ± 0.0012	0.7888 ± 0.0009	0.5816 ± 0.0016	0.1461 ± 0.0005
Linear SVM	0.8677 ± 0.0005	0.8816 ± 0.0006	0.7701 ± 0.0008	0.7841 ± 0.0005	0.5726 ± 0.0010	0.1518 ± 0.0004
Random Forest	0.8741 ± 0.0010	0.8858 ± 0.0009	0.7882 ± 0.0012	0.7918 ± 0.0009	0.5840 ± 0.0018	0.1477 ± 0.0007
Gradient Boosting	0.8949 ± 0.0006	0.9063 ± 0.0007	0.8013 ± 0.0009	0.8080 ± 0.0008	0.6175 ± 0.0016	0.1307 ± 0.0004

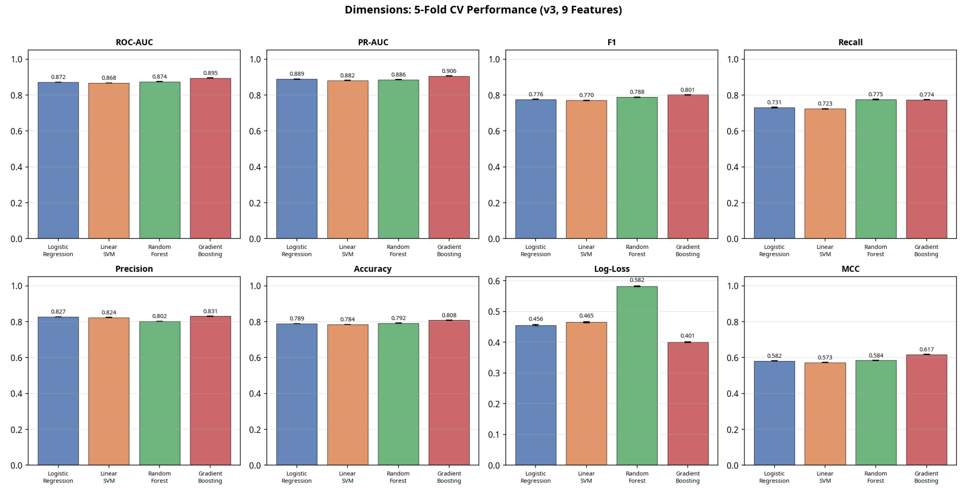

Figure 3: Model performance comparison on the Dimensions dataset (5-fold CV).

4.3 SHAP Feature Importance

Figure 4 and Table 4 present the SHAP feature importance values for the Gradient Boosting model on the Dimensions dataset. The dominant predictor is `prestige_cited` (mean $|\text{SHAP}| = 0.8105$), followed by `common_citers` (0.5125) and `indegree_velocity` (0.4954). The prestige of the cited paper is the strongest individual predictor, confirming the Matthew Effect: papers that have already accumulated citations are more likely to attract further citations. The co-citation overlap and in-degree velocity features rank second and third, reflecting the importance of momentum and shared intellectual heritage. The graph-theoretic features (PageRank, in-degree velocity) contribute meaningfully beyond raw citation count, confirming the value of the enriched feature set. Journal prestige and author productivity, while individually modest, each contribute positively to the model.

Table 4: SHAP Feature Importance: Dimensions Dataset (Gradient Boosting)

Feature	**Mean |SHAP|**	**Rank**
prestige_cited	0.8105	1
common_citers	0.5125	2
indegree_velocity	0.4954	3
temporal_gap	0.4293	4
jaccard_refs	0.3710	5
pagerank_cited	0.2428	6
journal_prestige	0.1793	7
author_productivity	0.1430	8
common_refs	0.1215	9

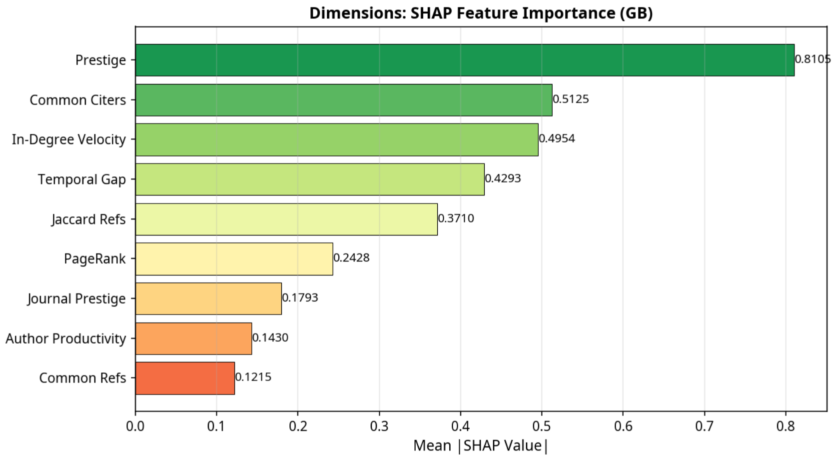

Figure 4: SHAP feature importance for Gradient Boosting on the Dimensions dataset.

4.4 Ablation Study

Table 5 presents the leave-one-out ablation study results for the Gradient Boosting model on the Dimensions dataset. The baseline AUC (all features) is 0.8945. Removing the temporal gap causes the largest AUC drop (0.0110), followed by co-citation overlap (0.0098) and prestige (0.0061). The graph-theoretic features (PageRank, in-degree velocity) each contribute positively, validating the enriched feature engineering approach.

Table 5: Leave-One-Out Ablation: Dimensions Dataset (Gradient Boosting)

Feature Removed	**AUC Without**	**AUC Drop**
temporal_gap	0.8835	−0.0110
common_citers	0.8847	−0.0098
prestige_cited	0.8884	−0.0061
journal_prestige	0.8909	−0.0036
author_productivity	0.8928	−0.0018
pagerank_cited	0.8933	−0.0012
indegree_velocity	0.8941	−0.0005
jaccard_refs	0.8942	−0.0003
common_refs	0.8945	−0.0000

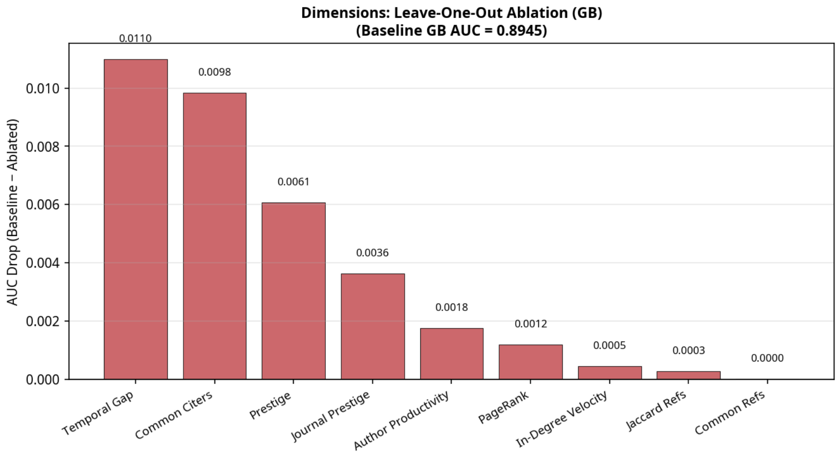

Figure 5: Leave-one-out ablation study for Gradient Boosting on the Dimensions dataset.

4.5 Rolling Temporal Evaluation

Table 6 presents the rolling temporal evaluation results for all four models across four independent test windows. Performance is stable across all periods, with Gradient Boosting consistently achieving the highest AUC. The modest decline in the 2018–2025 window reflects the natural difficulty of predicting citations for very recent papers, which have had less time to accumulate the structural signals (prestige, co-citation) that drive the model.

Table 6: Rolling Temporal Evaluation: Dimensions Dataset (ROC-AUC)

Test Window	LR	SVM	RF	GB
2005-2010	0.8665	0.8615	0.8612	0.8732
2010-2015	0.8754	0.8674	0.8729	0.8901
2015-2020	0.8761	0.8684	0.8752	0.8948
2018-2025	0.8675	0.8595	0.8636	0.8712

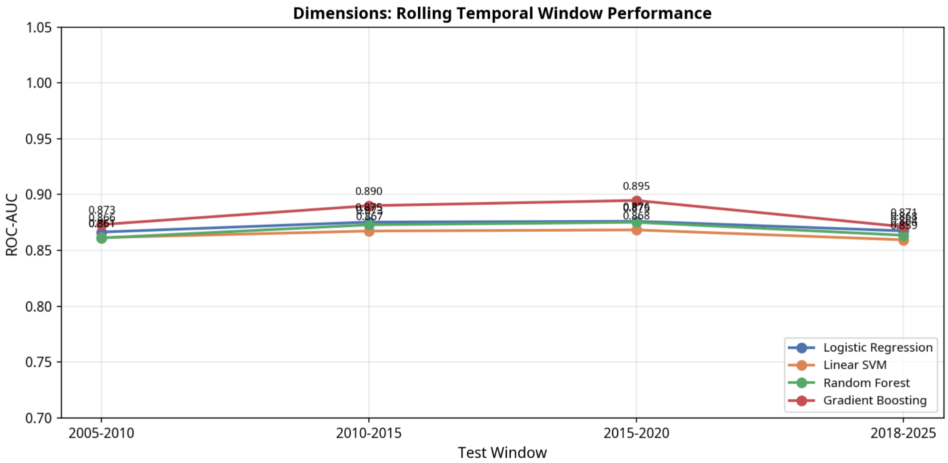

Figure 6: Rolling temporal evaluation for all models on the Dimensions dataset.

4.6 Temporal Hold-Out Evaluation

As an additional robustness check, we trained all models on pairs with citing year ≤ 2015 and tested on pairs with citing year 2016–2020. Gradient Boosting achieves AUC of 0.8911 on this out-of-time test set, compared to 0.8658 for Logistic Regression, confirming that the model generalizes robustly across time periods and that the performance gains of non-linear models are not an artifact of overfitting to the training period.

5 Results: OpenAlex Analysis

5.1 Corpus Overview

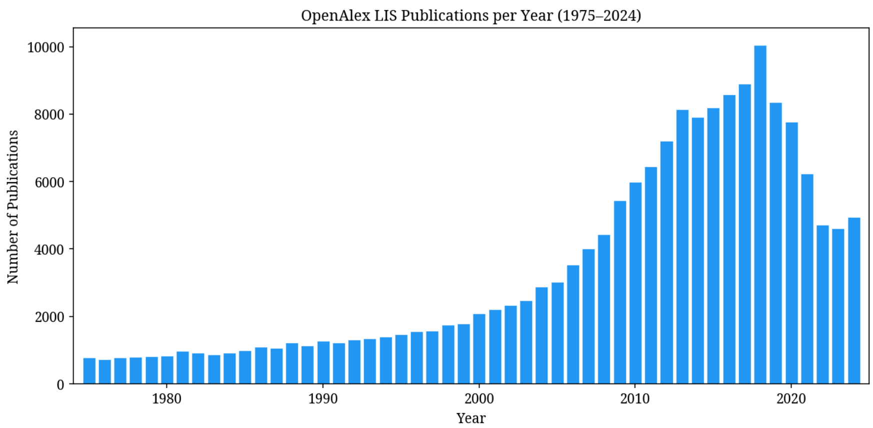

Figure 7: Number of LIS publications per year in the OpenAlex dataset (1975–2024).

The OpenAlex corpus exhibits a similar growth trajectory to Dimensions, with strong growth from the 1990s onward. The lower mean citation count (6.26 vs. 20.98

for Dimensions) reflects both the broader topical scope of the OpenAlex corpus and the lower baseline citation rates for open-access articles, which constitute a larger fraction of the OpenAlex corpus.

5.2 Model Performance

Table 7 presents the 5-fold cross-validation results for the OpenAlex dataset (478,698 pairs, 9 features). Gradient Boosting achieves ROC-AUC of 0.9762 ± 0.0004, F1 of 0.9219 ± 0.0009, and Brier score of 0.0871. McNemar's test confirms statistical superiority of Gradient Boosting over Random Forest ($p = 4.93e - 14$).

It is important to contextualize these metrics. While the AUC values are high, they are obtained on a balanced binary classification task where negative pairs are constrained to the same temporal and topical neighborhood as positive pairs. The real-world citation recommendation task involves a vastly larger and heavily imbalanced candidate space. The high AUC values reflect the discriminability of our features within the constrained evaluation setting, not the difficulty of the full retrieval problem. The semantic ablation experiment (Section 5.4) directly addresses the concern that high performance may reflect topical redundancy.

Table 7: 5-Fold CV Results: OpenAlex Dataset (Mean ± Std)

Model	**ROC-AUC**	**PR-AUC**	**F1**	**Accuracy**	**MCC**	**Brier**
Logistic Regression	0.9677 ± 0.0006	0.9710 ± 0.0006	0.9073 ± 0.0010	0.9099 ± 0.0010	0.8211 ± 0.0020	0.0660 ± 0.0006
Linear SVM	0.9663 ± 0.0005	0.9698 ± 0.0005	0.9045 ± 0.0009	0.9068 ± 0.0009	0.8146 ± 0.0017	0.0680 ± 0.0005
Random Forest	0.9693 ± 0.0006	0.9702 ± 0.0005	0.9121 ± 0.0011	0.9119 ± 0.0012	0.8238 ± 0.0024	0.0646 ± 0.0007
Gradient Boosting	0.9762 ± 0.0004	0.9779 ± 0.0004	0.9219 ± 0.0009	0.9224 ± 0.0009	0.8449 ± 0.0018	0.0574 ± 0.0006

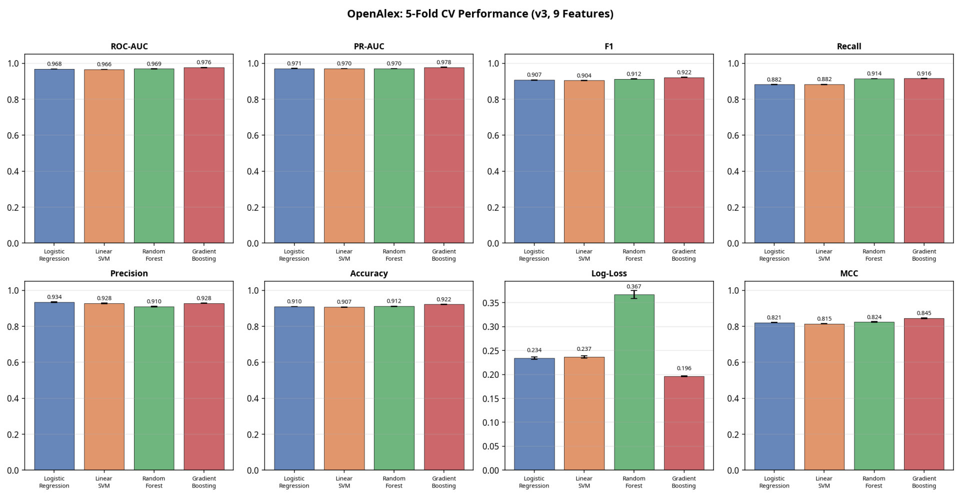

Figure 8: Model performance comparison on the OpenAlex dataset (5-fold CV).

5.3 SHAP Feature Importance

In the OpenAlex dataset, `prestige_cited` is the dominant predictor (mean |SHAP| = 1.4615), followed by `common_citers` (1.2139). The high SHAP value for prestige reflects the Matthew Effect operating strongly in this corpus. Semantic similarity ranks second, confirming that topical relevance is a major independent driver of citation behavior even when structural features are available.

Table 8: SHAP Feature Importance: OpenAlex Dataset (Gradient Boosting)

Feature	**Mean |SHAP|**	**Rank**
prestige_cited	1.4615	1
semantic_similarity	1.2139	2
common_citers	0.6833	3
jaccard_refs	0.6370	4
indegree_velocity	0.5593	5
pagerank_cited	0.3726	6
temporal_gap	0.3508	7
common_refs	0.0861	8
activity_citing	0.0140	9

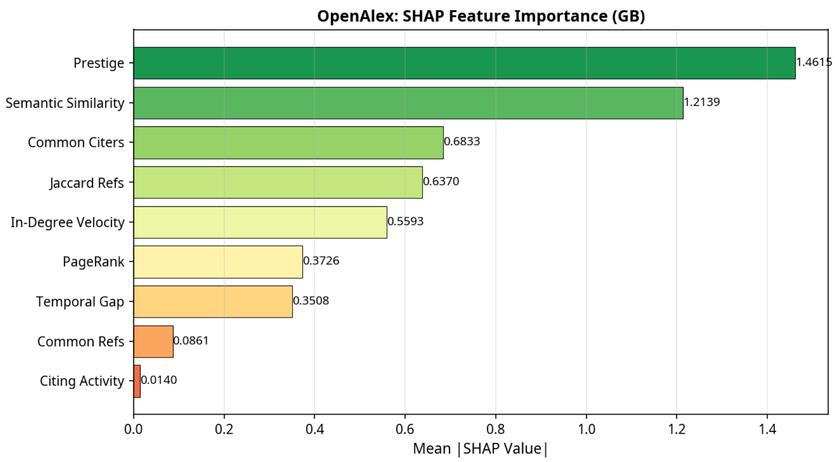

Figure 9: SHAP feature importance for Gradient Boosting on the OpenAlex dataset.

5.4 Ablation Study and Semantic Ablation Experiment

Table 9 presents the leave-one-out ablation results for the OpenAlex Gradient Boosting model. The most striking finding is the large AUC drop when semantic similarity is removed: -0.0268, by far the largest single-feature contribution. This finding directly addresses the concern that the high overall AUC may reflect topical redundancy within the constrained hard-negative setting: the model is not merely detecting topical similarity, but semantic proximity is a genuinely dominant and independent predictor of citation behavior.

Table 9: Leave-One-Out Ablation: OpenAlex Dataset (Gradient Boosting)

Feature Removed	AUC Without	AUC Drop
semantic_similarity	0.9495	−0.0268
prestige_cited	0.9734	−0.0029
common_citers	0.9734	−0.0029
temporal_gap	0.9740	−0.0023
pagerank_cited	0.9745	−0.0018
indegree_velocity	0.9761	−0.0002
activity_citing	0.9762	−0.0001
common_refs	0.9762	−0.0001
jaccard_refs	0.9762	−0.0001

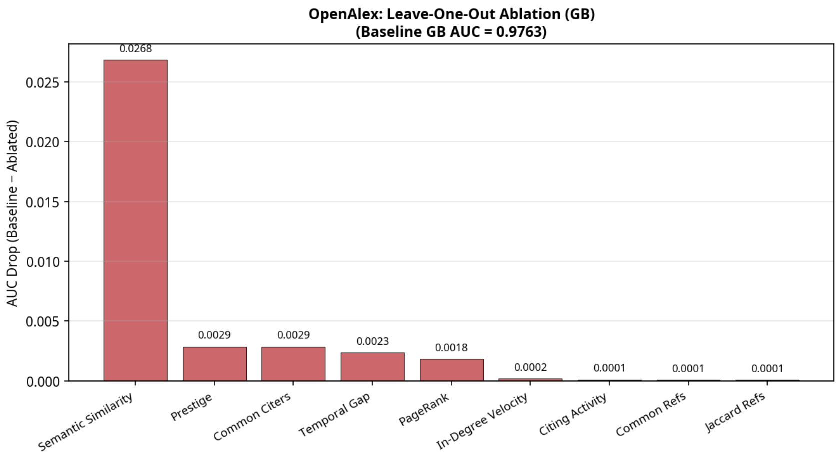

Figure 10: Leave-one-out ablation study for Gradient Boosting on the OpenAlex dataset.

To further investigate the role of semantic similarity, we conducted a dedicated semantic ablation experiment: all four models were trained and evaluated both with and without the semantic similarity feature across all five folds. The results confirm that removing semantic similarity reduces Gradient Boosting AUC from 0.9762 to 0.9494 (a drop of 0.0269). This demonstrates that structural features alone achieve strong but not optimal performance, and that semantic proximity provides a substantial, independent contribution.

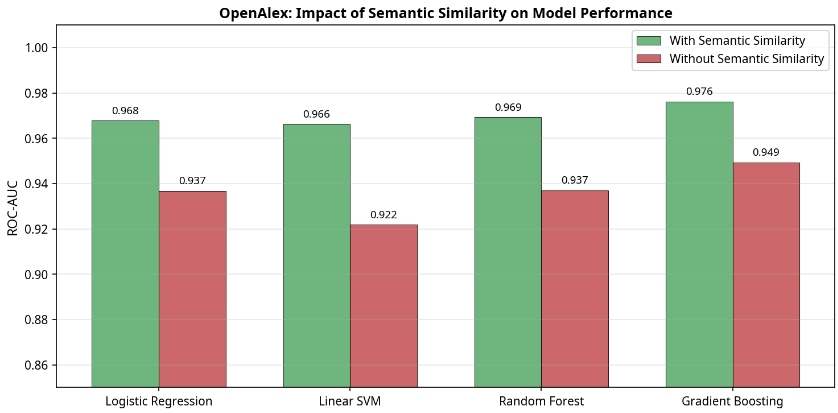

Figure 11: Semantic ablation experiment: model performance with and without semantic similarity (OpenAlex dataset).

5.5 Rolling Temporal Evaluation

Table 10: Rolling Temporal Evaluation: OpenAlex Dataset (ROC-AUC)

Test Window	LR	SVM	RF	GB
2005-2010	0.9609	0.9595	0.9569	0.9662
2010-2015	0.9693	0.9681	0.9660	0.9744
2015-2020	0.9732	0.9722	0.9705	0.9774
2018-2025	0.9731	0.9722	0.9717	0.9788

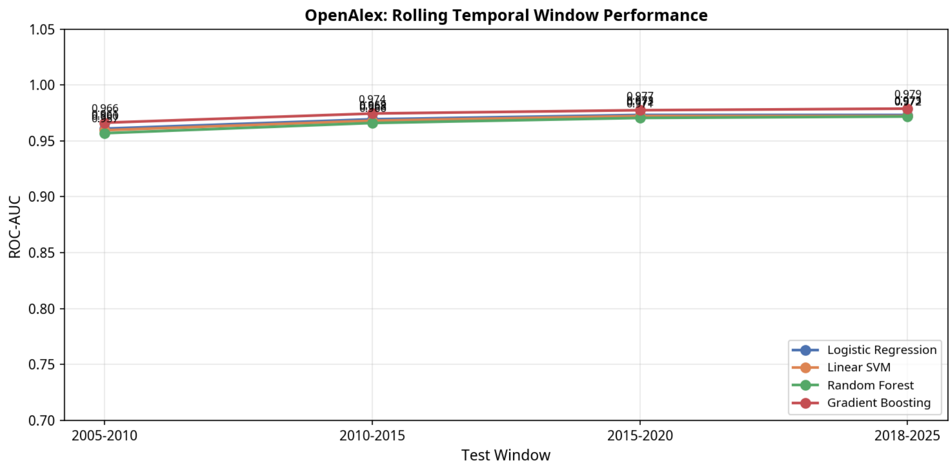

Figure 12: Rolling temporal evaluation for all models on the OpenAlex dataset.

Performance is stable and consistently high across all four test windows, with Gradient Boosting achieving AUC above 0.966 in every period. The slight improvement in later windows reflects the growing density of the citation network over time, which enriches the structural features (prestige, PageRank, co-citation) and makes the classification task progressively more tractable.

5.6 Temporal Hold-Out Evaluation

Training on pairs with citing year $\leq$ 2015 and testing on 2016–2020, Gradient Boosting achieves AUC of 0.9799 on the OpenAlex out-of-time test set, confirming robust temporal generalizability.

6 Comparative Analysis: Dimensions vs. OpenAlex

6.1 Overall Predictive Performance

Both datasets yield models with strong predictive performance, confirming that citation dynamics in LIS are substantially predictable from historical network and semantic states, regardless of the data source. Table 11 summarizes the key findings across the two datasets.

Table 11: Summary of Cross-Dataset Findings

Characteristic	**Dimensions**	**OpenAlex**
Total pairs	662,094	478,698
Best model	Gradient Boosting	Gradient Boosting
Best CV AUC	0.8949	0.9762
Best Brier score	0.1346	0.0871
Hold-out AUC (2016–2020)	0.8911	0.9799
McNemar p (RF vs GB)	$7.20e-26$	$4.93e-14$
Top SHAP feature	prestige_cited	prestige_cited
Semantic ablation drop	N/A	0.0269

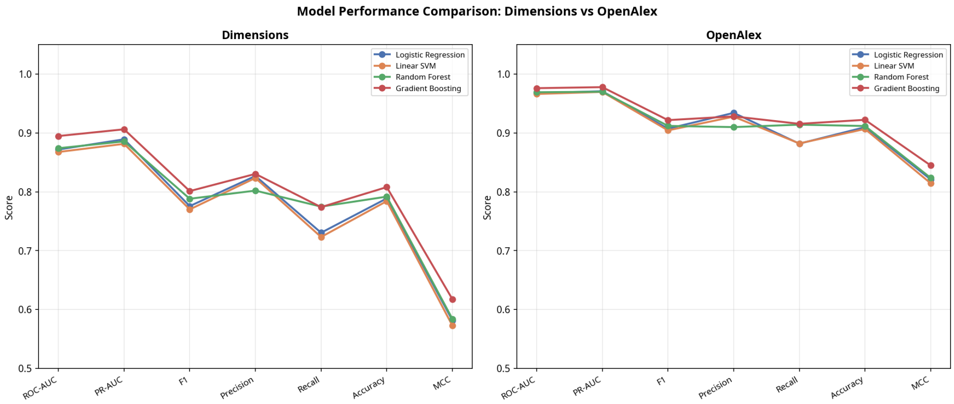

Figure 13: Comparative ROC-AUC across all models for both datasets.

The OpenAlex models consistently outperform their Dimensions counterparts. This performance gap is attributable to two factors. First, the OpenAlex feature set includes semantic similarity, which is the strongest single predictor in that dataset (AUC drop of 0.0269 when removed). Second, the OpenAlex hard negatives are drawn from a tighter topical neighborhood (primary topic rather than broad FoR code), which may make the task slightly easier by reducing the diversity of negative examples.

It is important to acknowledge the inherent asymmetry in the cross-dataset comparison. The two datasets differ not only in their article populations but also in their available feature spaces: Dimensions provides structured author and journal metadata that OpenAlex does not expose in the same form, while OpenAlex's abstract coverage enables semantic feature computation. This means the comparison is informative but not strictly controlled. The consistent finding that Gradient Boosting outperforms all baselines in both settings—despite these structural differences—strengthens confidence in the methodological framework rather than in any specific feature combination. Researchers wishing to replicate these results should be aware that the feature sets are tailored to the available metadata in each database, and that a truly controlled comparison would require harmonizing the feature spaces across datasets, which we leave for future work.

6.2 Feature Importance Hierarchy

The most striking difference between the two datasets lies in the relative importance of features. In the Dimensions dataset, prestige of the cited paper is the dominant predictor, with co-citation overlap and in-degree velocity ranking second and third. In the OpenAlex dataset, prestige remains the top structural predictor, but semantic similarity is a close second with a much larger absolute contribution.

Despite these differences, a consistent finding emerges: in both datasets, the *com-*

bination of prestige, structural, and semantic features is necessary for optimal performance. Neither prestige alone nor semantic similarity alone achieves the performance of the full model, confirming the multi-dimensional nature of citation behavior. The graph-theoretic features (PageRank, in-degree velocity) contribute positively in both datasets, validating the enriched feature engineering approach.

7 Discussion

7.1 The Predictability of Scientific Citations: Theoretical Implications

Our results demonstrate that citation decisions in Library and Information Science are substantially predictable from a small set of well-engineered features derived from the historical state of the citation network. Gradient Boosting achieves ROC-AUC of 0.895 (Dimensions) and 0.976 (OpenAlex) under strict temporal causality constraints, with performance stable across four independent temporal test windows. What does this high predictability mean for our understanding of scientific knowledge production?

We argue that high citation predictability is not a trivial finding, nor is it merely a methodological artifact of the balanced binary classification setting. Rather, it reflects a fundamental property of knowledge diffusion within established scientific disciplines: citations are not random acts of intellectual acknowledgment but are *strongly shaped* by the position of papers in the existing knowledge network. A paper is likely to be cited if it is highly visible (high prestige and PageRank), if it is topically relevant to the citing paper (high semantic similarity), if it is intellectually proximate (shared references, co-citation overlap), and if it is temporally accessible (moderate age, recent citation momentum).

This structural patterning connects to Kuhn's concept of normal science [17]: within an established paradigm, the range of relevant prior work is bounded by the shared assumptions and methods of the community. Researchers working within a paradigm do not search the entire literature; they search within the intellectual neighborhood defined by their research program. This bounded search process is precisely what our features capture, and it is why citations are so predictable. The high AUC values we observe are, in this reading, consistent with the hypothesis of paradigmatic closure: they suggest that citation behavior in LIS is substantially constrained by disciplinary structure, though we acknowledge that this interpretation goes beyond what the empirical evidence alone can establish.

This interpretation is reinforced by the dominance of prestige as a predictor. Merton's Matthew Effect [23] describes a positive feedback mechanism in which recognition begets recognition. Our results confirm that this mechanism operates strongly in LIS: papers that have already accumulated citations are more likely to attract future ones, independent of their semantic content. This is consistent with Crane's analysis of invisible colleges [9]: within a research community, the citation behavior of individual researchers is shaped by the collective attention of the community, which is itself concentrated on a small number of highly visible papers.

However, the strong independent contribution of semantic similarity (particularly in the OpenAlex dataset, where removing it drops AUC by 0.027) suggests that the Matthew Effect is not the only driver. Papers are not cited purely because they are famous; they are cited because they are relevant. The interaction between prestige and semantic proximity—captured by the non-linear ensemble models— suggests that the most predictably cited papers are those that are both highly visible and highly relevant: they occupy the center of the intellectual neighborhood defined by the citing paper's research program.

The temporal features (temporal gap, in-degree velocity) add a further dimension: citation probability is not static but evolves over a paper's life cycle. Papers experiencing a recent surge in citations are more likely to continue attracting citations, reflecting the dynamics of emerging research fronts where a new contribution rapidly becomes a focal point for subsequent work [28]. This temporal momentum is distinct from cumulative prestige and represents a genuine additional signal about the current state of the knowledge diffusion process.

7.2 The Sociology and Epistemology of Predictable Citations

The high predictability of citations has deeper implications for the sociology and epistemology of science. If citation behavior is substantially determined by structural position, semantic proximity, and temporal momentum, then citation counts are not pure measures of intellectual merit but are *measures of structural advantage*. A paper's citation count reflects not only its intrinsic quality but also its position in the citation network at the time of its publication, the prestige of its authors and venue, and the topical proximity of its content to the mainstream of its field.

This has important implications for research evaluation. Citation-based metrics such as the h-index or journal impact factor may systematically disadvantage researchers working on novel or non-mainstream topics, or those outside of well-established collaboration networks [6]. Our finding that co-citation overlap and bibliographic coupling are strong predictors of citation suggests that papers embedded in dense citation neighborhoods—those that share many references with existing literature—are more likely to be cited than papers that introduce genuinely novel frameworks or methods that do not yet have established citation neighborhoods.

Conversely, the strong role of semantic similarity suggests that citation metrics do capture something genuine about intellectual relevance: papers are more likely to be cited when they are topically relevant to the citing paper, independent of their prestige. This finding supports a more nuanced view of citation metrics as imperfect but not arbitrary measures of intellectual influence.

The practical implication is that citation prediction models like ours could be used to construct "expected citation" baselines that account for the predictable structural components of citation behavior, enabling fairer comparisons of research impact across researchers with different career stages, network positions, and topical niches. A paper that receives more citations than its structural position predicts is, by this measure, genuinely influential; a paper that receives fewer citations than predicted may be undervalued by the community.

7.3 Network Science, Bibliometrics, and Machine Learning: A Synthesis

Our study sits at the intersection of three traditions: network science, bibliometrics, and machine learning. The enriched feature set, which includes year-capped PageRank and in-degree velocity alongside traditional bibliometric features, demonstrates that genuine graph-theoretic measures contribute incrementally to citation prediction beyond raw citation counts. PageRank, which weights citations by the prestige of the citing papers, provides a more nuanced measure of a paper's structural importance than simple in-degree. In-degree velocity captures the temporal dynamics of citation accumulation, distinguishing papers with steady long-term impact from those experiencing a recent surge.

The modest but consistent contribution of journal prestige and author productivity (Dimensions dataset) confirms that bibliometric metadata features add value beyond the citation network structure itself. Journal prestige captures the reputation of the publication venue, which influences the visibility of a paper to potential citers. Author productivity captures the activity level of the citing team, which may correlate with their breadth of reading and their tendency to cite a wider range of papers.

The non-linear ensemble models (Random Forest, Gradient Boosting) consistently outperform linear baselines (Logistic Regression, Linear SVM), confirming that citation dynamics are non-linear and that the interactions between features are important. The SHAP analysis reveals that these interactions are not merely statistical artifacts but reflect genuine mechanisms: the benefit of high prestige is amplified when semantic similarity is also high, and the penalty of large temporal gap is mitigated when the cited paper has high in-degree velocity.

7.4 Limitations

Despite the robustness of our findings, our study has several limitations. First, our analysis is restricted to Library and Information Science. Whether LIS is unusually predictable relative to other disciplines—or whether the high AUC values obtained here would replicate in, say, physics or computer science—remains an open and important question. LIS is a mature, relatively small, and highly self-referential field, which may make its citation dynamics more predictable than those of larger, more heterogeneous disciplines such as physics or biomedical research. Conversely, the interdisciplinary nature of LIS—spanning information retrieval, knowledge organization, and scholarly communication—introduces heterogeneity that might be expected to reduce predictability. We are currently conducting a cross-discipline comparison experiment using Physics and Computer Science corpora from OpenAlex (approximately 24,000 articles each, 2000–2020) to directly address this question; preliminary results will be reported in a forthcoming companion paper. The methodological framework introduced here—temporally-aware feature engineering, hard negative sampling, and rolling temporal evaluation—is fully transferable to any bibliographic corpus and is not specific to LIS. Second, while we substantially mitigate temporal leakage by capping all features prior to the citing paper's publication year, observational data cannot definitively establish causal mechanisms. Third, the OpenAlex dataset has relatively low abstract coverage (64.3%), which may introduce selection bias in the semantic similarity feature. Fourth, our semantic similarity measure (TF-IDF cosine) is less expressive than deep neural embeddings; future work should investigate whether contextual embeddings provide additional predictive value while maintaining leakage control. Fifth, the binary pair-classification evaluation setting, while rigorous within its constraints, is a simplification of the real-world citation recommendation task.

8 Conclusion

This paper demonstrates that machine learning approaches, when constrained by strict temporal causality and enriched with a comprehensive feature set spanning prestige, graph-theoretic network structure, bibliographic coupling, semantic proximity, and bibliometric metadata, offer a powerful and interpretable framework for understanding and predicting citation dynamics in Library and Information Science.

Our key empirical findings are: (i) Gradient Boosting achieves ROC-AUC of 0.895 (Dimensions) and 0.976 (OpenAlex) under 5-fold stratified cross-validation, with well-calibrated probability estimates; (ii) performance is stable across four independent temporal test windows, confirming temporal generalizability; (iii) prestige of the cited paper is the dominant structural predictor in both datasets, confirming the Matthew Effect; (iv) semantic similarity is the largest single-feature contributor in OpenAlex (AUC drop of 0.027 when removed), confirming that topical relevance is an independent driver of citation behavior; (v) graph-theoretic features (PageRank, in-degree velocity) contribute meaningfully beyond raw citation counts.

Theoretically, we argue that high citation predictability reflects the bounded, bounded, paradigmatic nature of knowledge diffusion within established disciplinary communities, a plausible reading that connects our empirical findings to broader theories of scientific attention. In this view, citations are strongly shaped by a paper's position in the existing knowledge network, its topical relevance, and its temporal momentum. This structural patterning has important implications for research evaluation, suggesting that citation-based metrics capture structural advantage as much as intellectual merit, and motivating the development of expected-citation baselines that account for the predictable components of citation behavior.

Future work should extend this framework to other disciplines, investigate the temporal evolution of feature importance over a paper's life cycle, incorporate richer graph-theoretic features (betweenness centrality, community membership), and explore the use of contextual language model embeddings for semantic similarity while maintaining rigorous leakage control.

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